TV-D1 mission demonstrated an in-flight abort sequence in 2023. Five types of escape-system motors have since been developed and static-tested, while work towards the next test-vehicle mission, TV-D2, continues. Even the return to Earth is built around layers of redundancy.

Gaganyaan's deceleration system uses ten parachutes of four types. First, two parachutes remove the protective apex cover. Two drogues stabilise and decelerate the capsule. Three pilot parachutes then extract three mains. Importantly, only two of those three main parachutes are required for a safe landing.

This year, ISRO dropped a roughly 5.7-tonne simulated Crew Module from an Indian Air Force Chinook during IADT-02 and successfully ran the parachute sequence before recovery at sea. In July, another test qualified a main parachute against the maximum loads expected during G1.

It sounds almost mundane compared with a rocket launch. It is anything but. For astronauts coming home, pieces of fabric opening at the correct altitude and in the correct sequence may be among the most important machines on the spacecraft.

Vyommitra's useful emptiness

And then there is Vyommitra. The humanoid has inevitably become the friendly face of an uncrewed mission, but its real value is less theatrical. ISRO says Vyommitra can monitor module parameters, perform microgravity experiments and simulate human-like functions.

Think of it less as a robot astronaut and more as an instrumented passenger occupying the place where human activity will occur.

The larger spacecraft around it has to demonstrate the things humans will take for granted: cabin pressure, temperature and humidity control, avionics, communications, attitude control and safe de-orbit and re-entry. ISRO has realised an engineering model of the Environmental Control and Life Support System, whose job ultimately includes maintaining a habitable cabin environment.

Nobody applauds the life-support loop for being boring. But boring is precisely what you want.

Mission ends in the water

The final Gaganyaan technology is not actually in space. After re-entry, parachute descent and splashdown, the Crew Module must float correctly, remain recoverable and get its occupants out safely. ISRO and the Indian Navy have spent years rehearsing that unglamorous final act, from pool and harbour recovery trials to well-deck procedures.

In July, ISRO reported successful qualification testing of the Crew Module Uprighting System, designed to return the capsule to an upright position after splashdown. Recovery assets and a recovery plan have now been identified, according to the government.

That neatly captures why G1 matters. Gaganyaan is often described as India's human-spaceflight programme. For the next few months, it is better understood as India's human-safety programme. Every abort motor, valve, parachute, flotation bag, radio link and recovery procedure is there to remove one more unknown before astronauts climb aboard.



7 things Gaganyaan must prove before humans fly

1. **The launcher must become human-rated.** ISRO says development and ground testing of HLVM3's propulsion stages and structures are complete.
2. **The escape system must work when the rocket doesn't.** Five motor types have been developed and static-tested; TV-D2 remains part of the qualification campaign.
3. **The cabin must sustain an Earth-like environment.** An engineering model of Gaganyaan's Environmental Control and Life Support System has been realised.
4. **The spacecraft must manoeuvre, de-orbit and separate cleanly.** Crew- and Service-Module propulsion systems have been developed and qualified.
5. **Re-entry must keep the capsule stable** while its thermal protection system deals with atmospheric heating.
6. **Ten parachutes must slow a roughly 5.7-tonne module** enough for a survivable splashdown. Tests continued through 2026.
7. **Splashdown is not mission accomplished.** The module must remain upright, be located and be recovered quickly by teams led by the Indian Navy.